

F1

$$v^2 = v_0^2 - 2a\Delta x$$

$$2a\Delta x = v_0^2$$

$$a = \frac{v_0^2}{2\Delta x} = \frac{40^2}{2(200)}$$

$$a = 4 \left[\frac{m}{s^2} \right]$$

$$x = x_0 + v_0 t - \frac{1}{2} a t^2$$

$$200 = 40t - \frac{1}{2} (4) t^2$$

$$2t^2 - 40t + 200 = 0$$

$$t \rightarrow 10 [s]$$

F2

$$x = x_0 + v_{x0} t$$

$$x = v \cos \theta t_v$$

$$v_y = v_{y0} - g t$$

$$0 = v \sin \theta - g t_s$$

$$t_s = \frac{v \sin \theta}{g}$$

$$y = y_0 + v_{y0} t - \frac{1}{2} g t^2$$

$$H = v \sin \theta t_s - \frac{1}{2} g t_s^2$$

$$H = v \sin \theta \left(\frac{v \sin \theta}{g} \right) - \frac{1}{2} g \left(\frac{v \sin \theta}{g} \right)^2$$

$$H = \frac{v^2 \sin^2 \theta}{2g}$$

$$x = 4H$$

$$v \cos \theta t_v = 4 \left(\frac{v^2 \sin^2 \theta}{2g} \right)$$

$$\cos \theta t_s = \frac{2v \sin^2 \theta}{g}$$

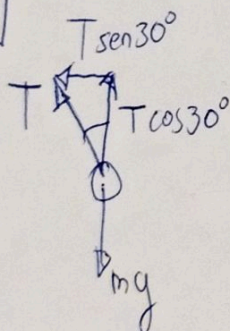
$$\cos \theta \left(\frac{v \sin \theta}{g} \right) = \frac{2v \sin^2 \theta}{g}$$

$$\cos \theta = \sin \theta$$

$$1 = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = 1 \rightarrow \theta = 45^\circ$$

F3



$$T \cos 30^\circ - mg = 0 \rightarrow T = \frac{mg}{\cos 30^\circ}$$

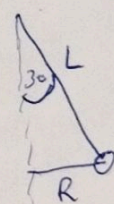
$$T \sin 30^\circ = m a_c$$

$$\frac{mg}{\cos 30^\circ} \sin 30^\circ = m \omega^2 R$$

$$g \tan 30^\circ = \omega^2 L \sin 30^\circ$$

$$\omega = \sqrt{\frac{g \tan 30^\circ}{L \sin 30^\circ}}$$

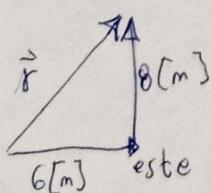
$$\omega = 2,38 \left[\frac{rad}{s} \right]$$



$$\sin 30^\circ = \frac{R}{L}$$

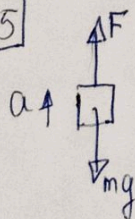
$$R = L \sin 30^\circ$$

F4



$$|\vec{r}| = \sqrt{8^2 + 6^2} \rightarrow |\vec{r}| = 10 [m]$$

F5



$$v^2 = v_0^2 + 2a\Delta x$$

$$2^2 = 2a(2)$$

$$a = 1 \left[\frac{m}{s^2} \right]$$

$$\sum F = ma$$

$$F - mg = ma$$

$$F = mg + ma$$

$$F = 33 [N]$$

$$W = F \cdot \Delta x$$

$$W = (33)(2)$$

$$W = 66 [J]$$